深度學習 HW3

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Homework File is in GitHub: https://github.com/PinChe567/deeplearning_homework/tree/HW3

1. Data and baseline analysis

The first thing I noticed was the data split. The supplied folder contains only 26 NORMAL images but 540 PNEUMONIA images. The original validation folder has 8 NORMAL images and no PNEUMONIA images, and the test folders are empty. With that split, a two-class validation AUROC or a final test result would not be meaningful.

Table 1. Image counts before and after splitting

Split	NORMAL	PNEUMONIA	Total
Provided train	18	540	558
Provided validation	8	0	8
Provided test	0	0	0
Rebuilt train	18	383	401
Rebuilt validation	4	62	66
Rebuilt test	4	95	99

I therefore combined the provided images and made a new split. PNEUMONIA images sharing the same personNNN token in the filename were kept in the same subset. NORMAL filenames did not provide a comparable patient ID, so each was treated as a separate group. This avoids some obvious overlap, but it does not prove that the sets are fully patient-independent. The new validation and test sets still contain only four NORMAL images each.

For repeated training on CPU, images were resized to 96 × 96 grayscale (original: 256 × 256). My reference model keeps the original idea of a three-hidden-layer MLP: 64 units per layer, BatchNorm, ReLU, and dropout 0.5. I trained it for six epochs (from 0 to 5) with Adam (learning rate 0.001), batch size 32, and weight decay 0.001. BCEWithLogitsLoss was used for the binary output. To reduce the class imbalance during training, a weighted sampler was applied to the training set only.

The best baseline checkpoint was at epoch 5: validation AUROC = 1.000 and balanced accuracy = 0.875. The AUROC improved markedly from epoch 1 to epoch 5, but the validation loss rose again at epoch 6 (0.187 → 0.252). I kept the checkpoint selected by validation AUROC instead of automatically using the last epoch.

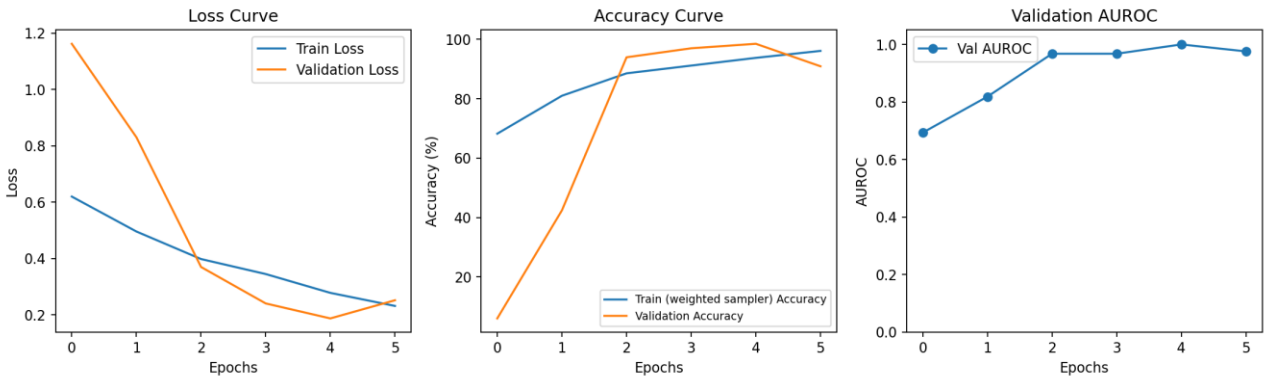


Figure 1. Training and validation curves for the baseline MLP. The training accuracy is measured on the weighted training sampler, whereas validation accuracy uses the natural class distribution.

2. Optimization settings

Below, I tested learning rate, batch size, and optimizer. Table 2 reports each run at the epoch with its highest validation AUROC.

Table 2. Optimization experiments

Setting	Best epoch	Val AUROC	Bal. acc.	Sensitivity	Specificity
Adam, LR 0.001 (baseline)	5	1.000	0.875	1.000	0.750
Adam, LR 0.0001	5	0.976	0.875	1.000	0.750
Adam, LR 0.01	5	0.980	0.952	0.903	1.000
Batch size 16	4	1.000	0.875	1.000	0.750
SGD + momentum, LR 0.01	5	0.996	0.750	1.000	0.500
AdamW, LR 0.001	5	0.996	0.750	1.000	0.500

The smaller learning rate did not improve AUROC in this short run. The larger rate gave the highest validation balanced accuracy (0.952), but its sensitivity dropped to 0.903. This is a useful reminder that a high AUROC does not mean the model will give the best classifications at a fixed 0.5 threshold. Reducing the batch size to 16 matched the baseline AUROC, although the best checkpoint occurred one epoch earlier.

AdamW and momentum SGD were close in AUROC, but both had lower balanced accuracy than the baseline at the selected checkpoint. The SGD comparison also uses LR 0.01 instead of 0.001, so it should not be interpreted as a perfectly controlled optimizer-only test. With one random seed and just four normal validation images, these differences are not strong evidence that one optimizer is generally better.

3. Training schedule and regularization

Table 3. Schedule and weight-decay experiments

Setting	Best epoch	Val AUROC	Bal. acc.	Specificity
Baseline: 6 epochs, wd = 0.001	5	1.000	0.875	0.750
12 epochs	5	1.000	0.875	0.750
Cosine LR schedule	6	0.976	0.859	0.750
No weight decay	5	0.996	0.750	0.500
Weight decay 0.05	5	1.000	0.500	0.000

Doubling the training budget from six to twelve epochs did not change the best selected checkpoint: it was still epoch 5. The training loss kept falling later, but the validation metrics did not improve consistently, so more epochs were not automatically helpful. Cosine scheduling produced a lower best validation AUROC (0.976) in this setup.

Removing weight decay slightly reduced AUROC. Raising it to 0.05 looked good if I checked AUROC alone (1.000), yet its specificity was zero at the 0.5 threshold—all four normal validation images were classified as pneumonia. That was one of the clearest examples of why I needed to look beyond a single metric. Also, because the training batches are deliberately resampled while

validation is not, the raw train–validation accuracy gap is not directly comparable; the losses and per-class measures give a more useful picture.

4. Model architecture

Table 4. Architecture experiments

Model	Best epoch	Val AUROC	Bal. acc.	Specificity
MLP, width 64, dropout 0.5	5	1.000	0.875	0.750
MLP, smaller width	5	0.964	0.750	0.500
MLP, lower dropout	4	1.000	0.750	0.500
Three-block CNN	4	0.911	0.500	0.000

I compared the original MLP with a narrower MLP, a lower-dropout version, and a simple three-block CNN. The CNN uses convolutional filters and global average pooling, so it has a stronger spatial bias than an MLP that flattens the image. Even so, it did not perform better here: its selected checkpoint reached AUROC 0.911 and classified every normal validation image as pneumonia.

The smaller MLP also underperformed the baseline. Lower dropout preserved AUROC 1.000 but reduced balanced accuracy from 0.875 to 0.750. I would not conclude that MLPs are better suited to chest X-rays based on this experiment alone. The models have different parameter counts, and six CPU-friendly epochs might not be enough for each architecture to reach its best performance.

5. Data augmentation

To isolate the augmentation comparison, I used the same CNN and training settings for all four conditions below. Only the training images were augmented; validation and test images were left unchanged.

Table 5. CNN augmentation experiments

Training transform	Best epoch	Val AUROC	Bal. acc.	Specificity
None	4	0.911	0.500	0.000
Mild rotation/shift/intensity	5	0.895	0.742	0.500
Horizontal flip	4	0.903	0.500	0.000
Strong geometric/intensity	6	0.935	0.500	0.000

The mild setting used rotations up to ±5°, translations up to 2%, and brightness/contrast changes up to ±8%. Strong augmentation increased these to ±20°, translations up to 12%, scale 0.8–1.2, and brightness/contrast changes up to ±30%. The flip condition reversed images left to right with probability 0.5. I included flipping as a comparison, not as a default recommendation for medical images.

Strong augmentation achieved the highest CNN AUROC (0.935), but its balanced accuracy stayed at 0.500 and specificity remained zero. Mild augmentation slightly lowered AUROC but improved balanced accuracy to 0.742. Neither observation is reliable enough to justify a clinical recommendation. Small positioning and intensity changes may be reasonable for some acquisition differences; large rotations, crops, or contrast changes can alter subtle opacities. Horizontal flipping is also questionable because it changes anatomical laterality and any left/right context in the image.

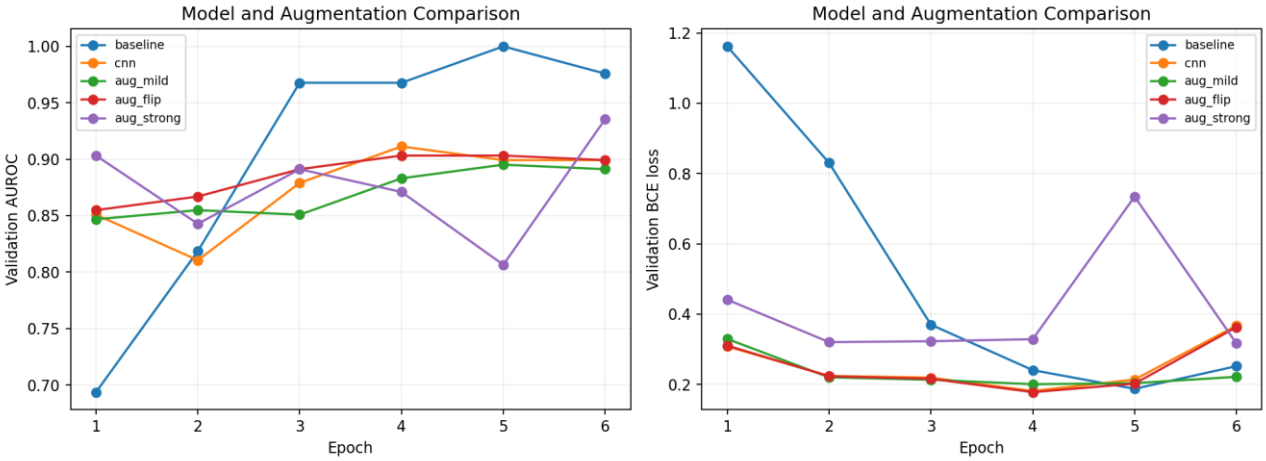


Figure 2. Validation AUROC (left) and BCE loss (right) across epochs for the baseline MLP, unaugmented CNN, and three CNN augmentation settings.

6. Conclusion

I selected the final checkpoint using validation AUROC, with validation balanced accuracy as the tie-breaker, and only then evaluated it on the held-out test split. The baseline MLP was selected from the 16 runs. Its test results are summarized below.

Table 6. Final baseline MLP test results (N = 99)

Accuracy	AUROC	Bal. acc.	Sensitivity	Specificity	Confusion matrix
97.98%	0.9921	0.750	1.000	0.500	TN 2 / FP 2 / FN 0 / TP 95

At first glance, 97.98% accuracy looks excellent. The confusion matrix changes that impression: 95 of the 99 test images were pneumonia, and two of the four normal images were false positives. The model detected all 95 pneumonia images at the chosen threshold, but the estimate of specificity rests on only four cases. I would report these numbers as results from this small assignment dataset, not as evidence of clinical diagnostic accuracy.

The baseline was the best choice within the comparison I ran, not necessarily the best architecture in general. The data limitation matters more than the small differences between many parameter settings. For a follow-up experiment, I would first collect substantially more normal images and confirm patient-level IDs, then repeat the comparisons with multiple random seeds and patient- or site-level holdouts. I would also allow enough training time for each model, report confidence intervals, and inspect errors before deciding on an operating threshold.

Complete experiment summary

Table 7 lists all 16 runs used for selection. “Best epoch” refers to the checkpoint with the highest validation AUROC, not necessarily the final training epoch.

Table 7. Validation results for all 16 runs

Run	Best epoch	AUROC	Bal. acc.	Sensitivity	Specificity
baseline	5	1.000	0.875	1.000	0.750
lr_low	5	0.976	0.875	1.000	0.750
lr_high	5	0.980	0.952	0.903	1.000
batch_16	4	1.000	0.875	1.000	0.750
optimizer_sgd	5	0.996	0.750	1.000	0.500

optimizer_adamw	5	0.996	0.750	1.000	0.500
epochs_12	5	1.000	0.875	1.000	0.750
schedule_cosine	6	0.976	0.859	0.968	0.750
weightdecay_0	5	0.996	0.750	1.000	0.500
weightdecay_high	5	1.000	0.500	1.000	0.000
mlp_small	5	0.964	0.750	1.000	0.500
mlp_drop_low	4	1.000	0.750	1.000	0.500
cnn	4	0.911	0.500	1.000	0.000
aug_mild	5	0.895	0.742	0.984	0.500
aug_flip	4	0.903	0.500	1.000	0.000
aug_strong	6	0.935	0.500	1.000	0.000

Appendix. Where the notebook was modified

The original 26 teacher cells remain in their original order in HW3_annotated_from_teacher.ipynb (zero-based cell numbering). Rather than replacing the whole notebook with unrelated code, the edited cells carry the changes and the additional runs are placed after cell 25.

Table 8. Report sections and corresponding original teacher cells

Report section	Teacher notebook cell(s)	Main change
Data / baseline	11, 14, 19, 22, 25	Fix split; set MLP; plot curves
Optimization	17, 19	Vary LR, batch size, optimizer
Schedule / regularization	17, 19	Vary epochs, scheduler, decay
Architecture	14, 19	Compare MLP variants and CNN
Augmentation	11, 19	Change training transforms
Conclusion / test	22, 25	Select on validation; evaluate test last

Important changes from the teacher example: the images are 96 × 96 instead of 256 × 256; the baseline uses six epochs, batch size 32, and a weighted training sampler; and the unusable supplied validation/test folders are replaced with grouped splits. The training functions and original MLP structure are still identifiable. Experiment configurations are added in cell 27, executed in cell 29, summarized in cell 30, and finally tested in cell 32.